

at five years, and these slightly stronger average effects arise simply from having more periods with a stable and fully realized effect.

C. Spillover Effects

A potential concern with our empirical strategy is whether displacement or spillover effects could be relevant to our main results. Online Appendix Figure A10 addresses this concern by comparing means for treated and nontreated units over event-times.²⁴ If spillover effects were a problem, we should see at least part of the effect on outcomes of interest coming from nontreated units breaking their trend when treated units receive CAPS. However, the changes in our outcomes during the postintervention period are concentrated in the treatment group's outcomes and we never see changes in the opposite direction in control areas. These results are consistent with the absence of displacement effects.

VI. Conclusion

The 2002 Brazilian psychiatric reform reorganized public mental health care delivery. A network of community-based services centered on CAPS was built across the country. To identify the causal effects of this reform, we exploit municipality-level variation in the establishment of CAPS. We find that the reform achieved its primary goal of increasing the supply of outpatient mental care and shifting care away from inpatient to outpatient settings. However, the effects are concentrated on long-stay admissions and patients with schizophrenia, an illness more severe than most CAPS are equipped to treat adequately. Despite community-care treatment being generally less expensive than inpatient treatment, we find evidence of increased public spending with the reform. Finally, we find a persistent increase in homicide rates following the reform, possibly caused by the loss of incapacitation effects that long-stay hospitalizations have on the crime-prone mentally ill.

To the best of our knowledge, this paper provides the first causal evaluation of a large-scale mental health reform. Our estimates are thus particularly informative for countries with deficits in providing mental health treatments (World Health Organization 2022), countries facing barriers to the implementation of a community-based strategy (e.g., due to financial constraints), and countries developing or revising deinstitutionalization interventions. According to the UN (2020), deinstitutionalization should be a priority following the COVID-19 pandemic.

Large-scale reforms that aim to change the nature of mental health care delivery, moving it toward community-based care, involve many aspects. Policymakers must be mindful of many factors that may affect mentally ill people. Our results from Brazil show that providing an inadequate substitute for more severe illnesses—in particular, schizophrenia—can generate undesirable effects for this population and the community in general. At the extreme, the negative consequences of failing to

²⁴To construct such time series, we first stack residualized (state $\times$ year fixed effects) long differences of the outcome (relative to event-time -1) for each treated cohort and its respective control group. We then collapse data at the event-time using weighted averages based on the treated cohort size.

offer necessary treatments can call into question the entire mental health care delivery strategy, regardless of its effectiveness when properly implemented.

A few details particular to our setting might offer important insights for future reforms in other contexts. The deinstitutionalization process in Brazil occurred late and still needs investment. Less than 3 percent of Brazilian municipalities have access to mental health centers open 24 hours a day, with outpatient beds and crisis intervention services. Such infrastructure is fundamental to reducing violence by people with serious mental illness in the community (Thornicroft and Tansella 2013; Lamb and Weinberger 2005; Dvoskin and Steadman 1994). Therefore, incorporating intensive care models into community settings can improve policy outcomes of mental health reforms. Evaluating these interventions is left to future work.

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